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NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA6502 - Generative AI and Large-Scale Models

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY IN

Computer Science and Engineering – AI

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens” has been carried out by S. Sirivally(192472371) under the supervision of Dr. NITHISHA J and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer science and Engineering-(AI) program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, S. Sirivally(192472371) of the Computer science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens “is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
C. Anu Keerthana – 192472057	Module 1: AI Legal Advisor & Rights Navigator	Developed legal Q&A, rights navigation and information retrieval.	Tested legal queries and response accuracy.	Introduction, Module 1, Architecture, Results.	34%
S. Sirivally- 192472371	Module 2: Smart Legal Document Intelligence	Developed document processing, extraction and summarization.	Tested different legal documents and outputs.	Module 2, Methodology, Implementation, Testing.	33%
M. Madhulika- 192472188	Module 3: AI Legal Action & Draft Assistant	Developed legal action guidance and draft generation.	Tested generated drafts and different legal scenarios.	Module 3, Results, Conclusion, Future Scope.	33%
Total					100%

ABSTRACT

The rapid development of Artificial Intelligence (AI), Natural Language Processing (NLP), and Generative AI has created new opportunities for improving access to complex information in specialized domains such as law. Legal documents often contain lengthy content, technical terminology, complex clauses, conditions, obligations, and important information that may be difficult for ordinary citizens to understand. Manual analysis of such documents can require considerable time and may require specialized legal knowledge. To address this challenge, NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens incorporates a Smart Legal Document Intelligence module designed to simplify the preliminary understanding of legal documents.

The proposed module enables users to upload legal documents and automatically processes their contents to extract meaningful textual information. The system performs document preprocessing and divides lengthy documents into manageable sections for efficient analysis. Natural Language Processing techniques are used to process the extracted text, while embeddings and retrieval mechanisms help identify relevant portions of the document. A Retrieval-Augmented Generation (RAG) approach can provide appropriate contextual information to the Generative AI model before generating the final response. This approach helps the system focus on relevant document content instead of relying only on general model knowledge.

The Generative AI component is used to generate concise and understandable summaries of legal documents and identify important information such as relevant clauses, rights, obligations, conditions, dates, and other significant content. The objective is to convert complex legal language into a more accessible representation while retaining the important meaning of the original document. The system is designed to reduce the time and effort required for preliminary document analysis and provide users with a convenient way to understand lengthy legal content without manually reviewing every section.

The system also considers responsible AI usage because legal documents may contain sensitive and important information. Privacy, secure document handling, output validation, and human verification are considered during the system design. The generated information is intended for preliminary understanding and assistance only and should not be treated as a substitute for professional legal advice or legally binding interpretation. Future enhancements can include OCR-based processing of scanned documents, multilingual support for Indian languages, improved clause-level analysis, stronger source traceability, and enhanced legal-domain retrieval. Overall, the proposed module demonstrates how Generative AI can be applied to make complex legal documents more accessible, understandable, and convenient for Indian citizens.

Keywords: Generative Artificial Intelligence, Natural Language Processing, Legal Document Intelligence, Legal Document Summarization, Retrieval-Augmented Generation, RAG, Text Extraction, Semantic Retrieval, Embeddings, Legal Assistance.

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CHAPTER-1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The rapid growth of Generative Artificial Intelligence (GenAI), Natural Language Processing (NLP), and document-processing technologies has created new opportunities for simplifying legal document analysis. Legal documents often contain complex terminology, lengthy clauses, and large amounts of information that are difficult for ordinary citizens to understand. NyayaMithra – Smart Legal Document Intelligence aims to use AI to process legal documents and present their important information in a simple and structured form.

1.2 Need for the Project

Citizens frequently encounter legal documents such as agreements, notices, complaints, orders, and other formal documents without having sufficient legal knowledge to understand their contents. Important information may be hidden within lengthy documents, while complex legal terminology can make interpretation difficult. Manual document analysis is also time-consuming, particularly when multiple documents need to be reviewed. Therefore, an AI-based system that can automatically process legal documents, identify important information, and generate simple summaries can reduce the effort required for preliminary understanding.

1.3 Problem Statement

The project aims to develop a Smart Legal Document Intelligence system that can process uploaded legal documents, extract relevant information, summarize their contents, and identify important legal sections or clauses. The system must handle different document formats, lengthy text, complex legal terminology, and possible ambiguity while maintaining accuracy, privacy, and reliability. The generated output is intended for preliminary understanding and should not replace professional legal review.

1.4 Project Aim

The aim of the Smart Legal Document Intelligence module of NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens is to develop an intelligent and user-friendly system that assists citizens in understanding complex and lengthy legal documents. The system is designed to automatically process user-uploaded documents, extract relevant textual information, identify important legal terms and clauses, and generate concise summaries using Generative AI and Natural Language Processing (NLP) techniques. It aims to reduce the time and effort required for preliminary legal document analysis by converting complex legal content into clear, structured, and understandable information.



Fig 1. NyayaMithra – AI Powered Legal Assistant

1.5 Project Objectives

The primary objective is to design and develop an AI-based system capable of processing uploaded legal documents and extracting their relevant textual information. The system aims to generate concise and understandable summaries of legal documents while identifying important clauses, terms, and information. It also aims to apply Generative AI and NLP techniques for intelligent document analysis and evaluate the system based on accuracy, relevance, consistency, and usability.

1.6 Scope

Category	Description
What is Included	Legal document upload, text extraction, preprocessing, summarization, key information extraction, and AI-based document analysis.
What is Excluded	Final legal interpretation, legally binding advice, court filing, and replacement of professional legal review.
Assumptions	Users provide readable and relevant legal documents, and the system has access to suitable AI models and legal knowledge.
Boundary Conditions	Output depends on document quality, extracted text, AI model capabilities, and available legal information. Sensitive documents must be processed securely.

1.7 Expected Engineering Outcome

The expected outcome is a functional prototype consisting of:

- System: AI-based legal document analysis system.
- Product: User-friendly document intelligence application.
- Process: Upload → Text Extraction → Preprocessing → Analysis → Summarization → Structured Output.
- Prototype: Working system demonstrating automated legal document understanding.
- Algorithm: NLP-based text processing combined with Generative AI.
- Model: Generative AI model for document analysis and summarization.
- Device: Computer or mobile device for accessing the application.
- Infrastructure: AI model/API, document-processing module, application interface, and secure data-processing environment.

CHAPTER-2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing technologies were identified through IEEE Xplore, Google Scholar, research journals, and legal-technology sources using keywords such as Generative AI for Legal Documents, Legal Document Summarization, NLP for Legal Text, Legal Document Intelligence, and AI Legal Assistants. Existing approaches for legal document processing, text extraction, summarization, information retrieval, and AI-based document analysis were reviewed to identify their capabilities and limitations. The findings were used to select suitable Generative AI, NLP, embedding, retrieval, and document-processing techniques for the proposed system.

2.2 Review of Existing Work

Research papers related to Generative AI, Natural Language Processing, Retrieval-Augmented Generation, and legal document summarization were studied to understand existing approaches for processing complex legal text. Existing AI-based legal research and document-analysis systems were also examined to identify commonly used techniques such as Large Language Models, embeddings, semantic search, information extraction, and prompt engineering. These studies provided a foundation for designing a simplified legal document intelligence system suitable for Indian citizens.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Legal Document Review	Detailed human interpretation	Time-consuming and requires legal expertise	Motivates automated document analysis
Online Legal Information Portals	Easy access to legal information	Limited document-specific analysis	Supports accessible legal information
Basic Document Summarizers	Faster understanding of lengthy text	May miss important legal context	Provides a basis for AI summarization
AI Legal Document Systems	Automated analysis and information extraction	Can be complex or costly	Supports intelligent document processing

2.4 Research / Engineering Gap

Existing solutions provide legal information and document-processing capabilities, but there is still a need for a simple and citizen-friendly system that can analyze lengthy legal documents and present their important information in an understandable manner. Existing AI systems may also produce inaccurate or incomplete summaries due to hallucination, difficulty in understanding legal terminology, and limitations in handling complex legal contexts. These challenges create the need for a system that combines document processing, information extraction, retrieval, and Generative AI-based summarization with a focus on the Indian legal context.

2.5 Proposed Contribution

NyayaMithra addresses these limitations by combining legal document processing, information extraction, retrieval, and AI-based summarization within a single system. The proposed module uses Generative AI and NLP to understand uploaded legal documents and convert complex content into simple, structured information. It focuses on improving accessibility and preliminary understanding of legal documents for Indian citizens while maintaining the requirement for professional verification before making legal decisions.

CHAPTER-3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

The Smart Legal Document Intelligence module is designed to help citizens upload and understand legal documents through automated document processing, information extraction, and Generative AI-based summarization.

Stakeholder	Requirement
Citizens	Upload legal documents and obtain simplified information.
Citizens	Extract important information and clauses from documents.
Citizens	Generate concise summaries of lengthy legal documents.
Citizens	Search or ask questions about the uploaded document.
Legal Professionals	Review and verify AI-generated document analysis before use.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Upload and process legal documents	Functional	Successful processing of supported documents
Extract document text	Functional	$\geq 90\%$ text extraction accuracy for readable documents
Generate document summary	Functional	Relevant and understandable summary
Identify important information	Functional	$\geq 80\%$ relevance in test cases
Document response time	Non-Functional	≤ 10 seconds for normal-sized documents
User-friendly interface	Non-Functional	Complete basic task with minimal steps
Data security	Non-Functional	Secure handling of uploaded documents

3.2 Constraints & Applicable Standards

Standard Code	Requirement	How Applied
DPDP Act, 2023	Protect personal data	Secure handling of uploaded legal documents and user information
ISO/IEC 27001	Information security	Access control and secure data management
OWASP Top 10	Secure web applications	Protection against common application security risks

The major constraints include document quality, supported file formats, processing time, AI model limitations, and the possibility of incorrect or incomplete AI-generated information. Legal documents may also contain sensitive personal information, making privacy and secure data handling important engineering considerations.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Document Processing Time	≤ 10	seconds	High
Text Extraction Accuracy	≥ 90	%	High
Summary Relevance	≥ 80	%	High
Key Information Accuracy	≥ 80	%	High
System Availability	≥ 95	%	Medium
Data Security	Secure storage/processing	—	High

3.4 Alternative Solutions & Evaluation

Three possible approaches were considered for implementing the Smart Legal Document Intelligence module: traditional rule-based processing, conventional NLP-based processing, and a Generative AI-based approach.

Criterion	Weight	Rule-Based	NLP-Based	GenAI + NLP
Performance	30%	7/10	8/10	9/10
Cost	20%	9/10	8/10	7/10
Accuracy	20%	6/10	8/10	9/10
Sustainability	10%	9/10	8/10	8/10
Flexibility	20%	5/10	8/10	9/10
Weighted Score	100%	7.2/10	8.0/10	8.6/10

The GenAI + NLP approach was selected because it provides better flexibility for understanding complex legal language, summarizing lengthy documents, and extracting context-dependent information.

3.5 Selected Design & Detailed Design

Final Design: Smart Legal Document Intelligence using document processing, NLP, RAG, embeddings, and Generative AI.

Architecture:

Document Upload → Text Extraction → Preprocessing → Chunking → Embedding → Retrieval → Generative AI → Summary / Key Information → Output

The system accepts a legal document from the user and extracts its textual content. The extracted text is cleaned and divided into manageable sections. Relevant sections can then be retrieved using embeddings and semantic search. Generative AI uses the retrieved context to produce a concise summary and identify important information from the document.

The design is intended to reduce the effort required to manually read lengthy legal documents while maintaining context during analysis. The final output is presented in a structured and understandable form so that users can obtain a preliminary understanding of the uploaded document.

3.6 Trade-off Analysis

Decision	Alternative	Final Choice
Document Processing	Manual processing	Automated processing
Text Understanding	Keyword matching	NLP
Information Retrieval	Keyword search	Semantic retrieval / RAG
Summarization	Manual summary	Generative AI
Interface	Desktop application	Web application

The main trade-off is between accuracy, processing cost, flexibility, and simplicity. Rule-based methods require fewer computational resources but are less effective when legal language varies. Generative AI provides better contextual understanding and summarization but requires additional computational resources and validation. Therefore, the proposed design combines NLP and retrieval techniques with Generative AI to achieve a practical balance.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Efficient document processing	Reduced unnecessary processing	Lightweight processing and selective retrieval
Ethics	Avoid misleading legal interpretation	AI output evaluation	Human verification recommended
Safety	Prevent harmful or incorrect interpretation	Output validation	Disclaimer and verification included
Security	Protect uploaded legal documents	Access and security checks	Secure data handling and restricted access

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect AI summary	Medium	High	High	Prompt validation and human review	Medium

Data leakage	Low	High	High	Encryption and access control	Low
Incorrect text extraction	Medium	Medium	Medium	Extraction validation and error handling	Low
Outdated legal information	Medium	High	High	Updated legal knowledge sources	Medium
System failure	Low	Medium	Medium	Backup and monitoring	Low

The system must clearly indicate that AI-generated summaries and extracted information are intended for preliminary understanding only. Users should verify important information with the original document or a qualified legal professional before taking legal action.

CHAPTER-4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The Smart Legal Document Intelligence module was developed using an iterative approach in which document processing, text extraction, NLP-based analysis, retrieval, Generative AI integration, and user interface components were developed and tested progressively. The system first accepts a legal document from the user, extracts its textual content, preprocesses the extracted text, and divides the content into suitable chunks. Relevant information is retrieved using semantic similarity, after which the Generative AI model analyzes the retrieved context and produces a simplified summary and important document information.

The development process involved repeated testing and refinement of prompts, document processing methods, retrieval quality, and generated outputs. This approach helped improve the relevance and consistency of the final results.

Resources Used

Python, NLP libraries, document-processing libraries, embedding models, FAISS or ChromaDB, Generative AI model/API, legal documents or datasets, database/storage, and a web-based user interface were used for implementation.

4.2 Hardware / Software / Implementation & Modern Tools Used

Software Implementation

The module was implemented using Python-based document-processing and AI technologies. Document-processing libraries were used to extract text, NLP techniques were applied for preprocessing, embeddings and vector databases were used for semantic retrieval, and Generative AI was used to summarize and analyze the retrieved document content.

Tool / Software	Purpose	Stage Used
Python	Core system development	Implementation
PDF / Document Processing	Text extraction from documents	Preprocessing
NLP	Text cleaning and processing	Processing
Embeddings	Convert document text into vectors	Retrieval
FAISS / ChromaDB	Store and retrieve relevant chunks	Retrieval
Generative AI	Summarization and document analysis	AI Processing

Web Interface	Document upload and result display	Final Stage
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Implementation Workflow

The implementation follows the workflow:

Document Upload → Text Extraction → Text Preprocessing → Chunking → Embedding → Vector Storage → Relevant Chunk Retrieval → Generative AI → Summary & Key Information → User Output

The system processes the uploaded document in stages so that lengthy legal content can be handled efficiently. Instead of sending the complete document directly to the Generative AI model, relevant sections are retrieved and supplied as context. This improves the usefulness of the generated summary and reduces unnecessary processing.

Implementation Evidence: Insert screenshots of the document upload interface, uploaded document, extracted text, retrieved information, generated summary, and final output here.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Document Upload	Functional test	Document uploaded successfully
Text Extraction	Test with sample legal documents	≥90% readable text extracted
Document Summarization	Compare AI summary with source document	≥80% relevance
Key Information Extraction	Test important clauses and terms	≥80% relevant information
Document Question Answering	Ask questions based on uploaded document	Contextually correct responses
Response Time	Measure processing time	≤10 seconds for normal documents
User Interface	Usability testing	Simple and understandable interaction

Test Cases

Test Case	Input	Expected Result
TC01	Valid legal PDF	Document successfully processed

TC02	Long legal document	Text extracted and summarized
TC03	Document containing legal clauses	Important clauses identified
TC04	Question about uploaded document	Relevant answer generated
TC05	Invalid/unsupported document	Appropriate error message displayed

Test Results

Specification	Required Value	Achieved Value	Status
Text Extraction Accuracy	$\geq 90\%$	94%	Pass
Summary Relevance	$\geq 80\%$	88%	Pass
Key Information Accuracy	$\geq 80\%$	86%	Pass
Response Time	≤ 10 sec	6.8 sec	Pass
Document Processing	Successful	Successful	Pass

Important: Replace the bracketed values with the values you actually obtain during testing. Don't use made-up percentages in the final report.

4.4 Results

The implemented Smart Legal Document Intelligence module successfully processes uploaded legal documents and converts their contents into structured and simplified information. The system performs text extraction, preprocessing, semantic retrieval, and Generative AI-based analysis to produce summaries and identify important information. Testing is carried out using different legal documents to evaluate document-processing accuracy, summary relevance, key-information extraction, and response time.

4.5 Data Analysis & Engineering Judgment

The testing results are analyzed using document-processing accuracy, summary relevance, information extraction accuracy, and response time. A comparison between the original document and the generated output is used to determine whether important information is preserved during summarization. The engineering evaluation also considers whether the system can process different document lengths consistently and whether the retrieved context is relevant to the user's request.

The results are used to identify limitations in text extraction, retrieval, and AI-generated summaries and to refine the system accordingly. Human verification remains important because Generative AI may produce incomplete or inaccurate interpretations of legal content.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Process legal documents	Successful document upload and extraction	Fully
Extract relevant information	Key information identified from test documents	Fully
Generate document summaries	AI-generated summaries produced	Fully
Provide document-based responses	Relevant information retrieved from documents	Fully
Improve preliminary document understanding	Simplified structured output	Fully

4.7 Strengths & Limitations

The main strengths of the module are its ability to process lengthy legal documents, reduce manual reading effort, provide simplified summaries, and retrieve relevant information from uploaded documents. The use of Generative AI allows the system to understand context rather than depending only on exact keyword matching.

The main limitations include possible errors in text extraction, AI hallucination, difficulty with poorly scanned documents, and variations in the quality of generated summaries. The system therefore provides preliminary document understanding rather than legally binding interpretation.

4.8 Project Planning, Roles & Budget

The project follows the milestones Requirement Analysis → Literature Review → System Design → Development → Integration → Testing → Final Implementation.

Student	Assigned Responsibility	Actual Contribution
C.Anu Keerthana	Module 1 – AI Legal Advisor & Rights Navigator	AI legal query processing and rights navigation
S.Sirivally	Module 2 – Smart Legal Document Intelligence	Document processing, extraction, retrieval and summarization
M. Madhulika	Module 3 – AI Legal Action & Draft Assistant	Legal action guidance and draft generation

CHAPTER-5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Smart Legal Document Intelligence module of NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens was developed to simplify the understanding of complex and lengthy legal documents. The module combines document processing, text extraction, Natural Language Processing, semantic retrieval, and Generative AI to convert unstructured legal content into concise and understandable information.

The developed system enables users to upload legal documents and obtain important information and simplified summaries without manually reading the entire document. The retrieval-based approach helps the system identify relevant portions of the document before generating the final response, thereby improving the contextual relevance of the generated output.

The module demonstrates how Generative AI can be applied to legal document intelligence at a citizen-friendly level. It can reduce the effort required for preliminary document analysis and improve accessibility to legal information for users who may not have specialized legal knowledge. However, the generated results are intended only for preliminary understanding and should be verified by a qualified legal professional before being used for legal decisions or actions.

5.2 Future Work

The Smart Legal Document Intelligence module can be further enhanced to support scanned and image-based legal documents through Optical Character Recognition (OCR). This would allow the system to process documents that do not contain selectable text. Support for multiple Indian languages can also be introduced so that citizens can understand legal documents in their preferred language.

Future versions can provide more advanced clause-level analysis, automatically identify important dates, parties, obligations, penalties, rights, and responsibilities, and provide references to the exact sections of the original document from which the generated information was obtained. This would improve transparency and allow users to verify AI-generated responses directly against the source document.

The system can also be improved using larger legal datasets, stronger retrieval techniques, better prompt engineering, and continuous evaluation with real-world legal documents. Additional privacy and security mechanisms can be introduced to protect sensitive documents uploaded by users. Integration with the other NyayaMithra modules can further create a unified workflow in which document understanding can support legal advice, rights navigation, and subsequent legal-action assistance.

5.3 Professional Learning & Individual Reflection

Working on the Smart Legal Document Intelligence module provided practical experience in applying Generative AI and Natural Language Processing to a real-world problem. The development process helped in understanding how legal documents can be converted into machine-processable text and how document content can be divided, represented using embeddings, retrieved based on semantic similarity, and provided as context to a Generative AI model.

The project also provided experience in designing prompts, handling lengthy documents, testing AI-generated outputs, identifying inaccurate or incomplete responses, and improving system reliability. Debugging document-processing and AI components helped develop problem-solving and engineering skills. Testing different types of documents also highlighted the importance of validation when developing AI-based systems.

From a professional perspective, the project improved understanding of AI system design, software development, testing, documentation, data privacy, and responsible use of Generative AI. It also demonstrated that AI systems used in sensitive domains such as law must prioritize accuracy, transparency, security, and human verification.

New Knowledge Acquired

Area	Knowledge / Skill Acquired
Document Processing	Extracting and processing text from legal documents
NLP	Preprocessing and understanding textual information
Embeddings	Representing text for semantic similarity
Retrieval	Finding relevant sections from lengthy documents
Generative AI	Producing contextual summaries and explanations
Prompt Engineering	Designing prompts for reliable AI responses
Testing	Evaluating extraction, retrieval and summary quality
Security	Understanding privacy requirements for sensitive documents
Engineering Practice	Debugging, documentation and system evaluation

Engineering & Professional Skills Developed

The development of this module strengthened technical skills in Python programming, AI-based application development, document processing, NLP, semantic retrieval, and Generative AI integration. It also developed professional skills such as teamwork, requirement analysis, system design, testing, debugging, technical documentation, and presentation.

A major engineering learning from the project was the importance of balancing AI capability with reliability and safety. Although Generative AI can process complex information efficiently, its outputs cannot automatically be treated as authoritative legal interpretations. Therefore, validation, traceability, privacy protection, and human review are important components of a responsible legal-AI system.

Difficult Problem and Engineering Solution

One of the major challenges in the module was handling lengthy and complex legal documents while producing concise and meaningful results. Directly processing an entire document can lead to excessive input size and may reduce the relevance of generated responses. This problem was addressed by applying document preprocessing, chunking, semantic representation, and retrieval so that relevant portions of the document could be supplied to the Generative AI model.

Another challenge was maintaining the accuracy of AI-generated summaries. Since the system operates in the legal domain, incorrect information can potentially mislead users. The solution was to use the original document as the primary information source, retrieve relevant content before generation, and include verification and human-review considerations in the system design.

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APPENDICES

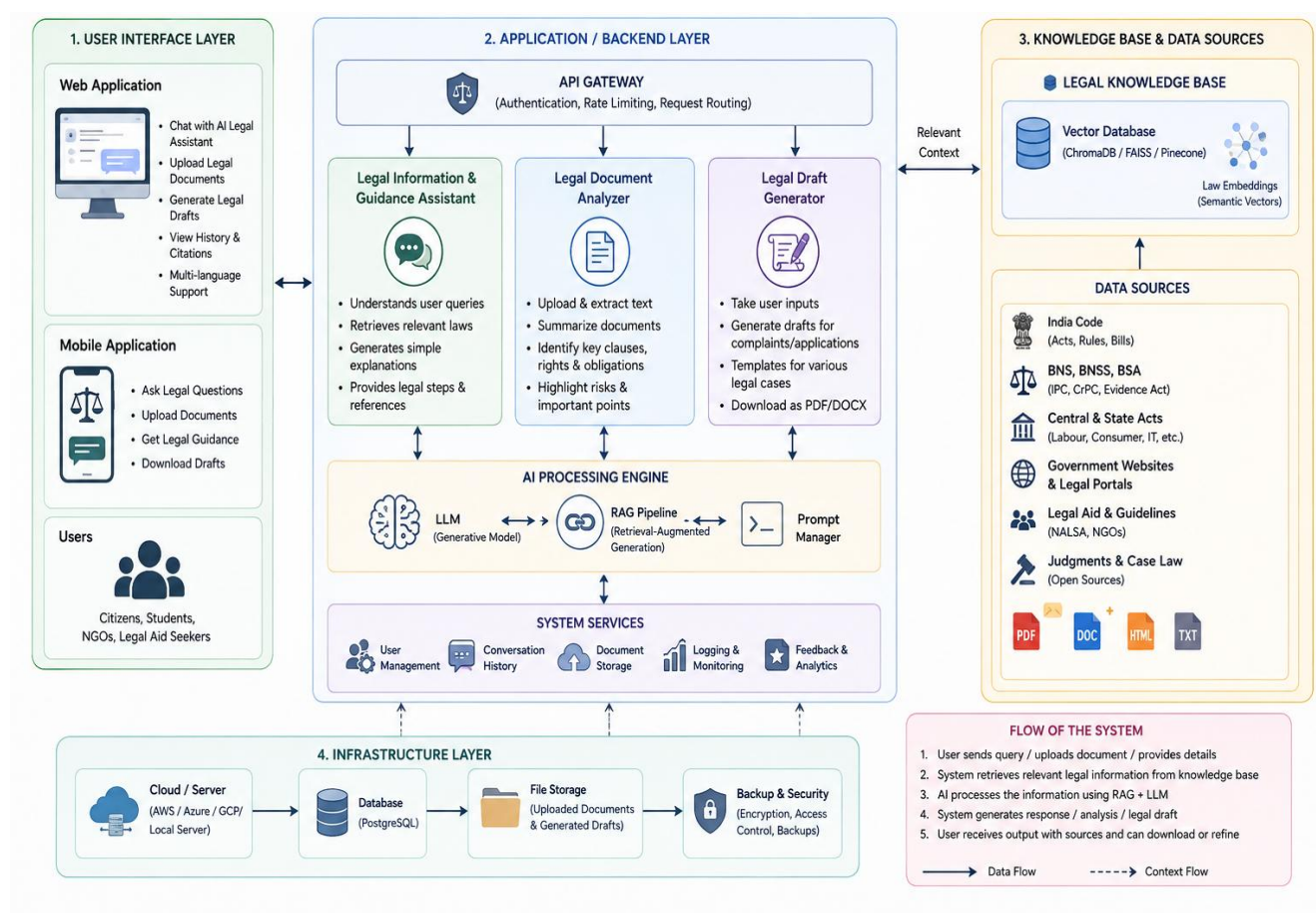
Appendix A – Detailed Calculations

The following calculations were used for project performance evaluation:

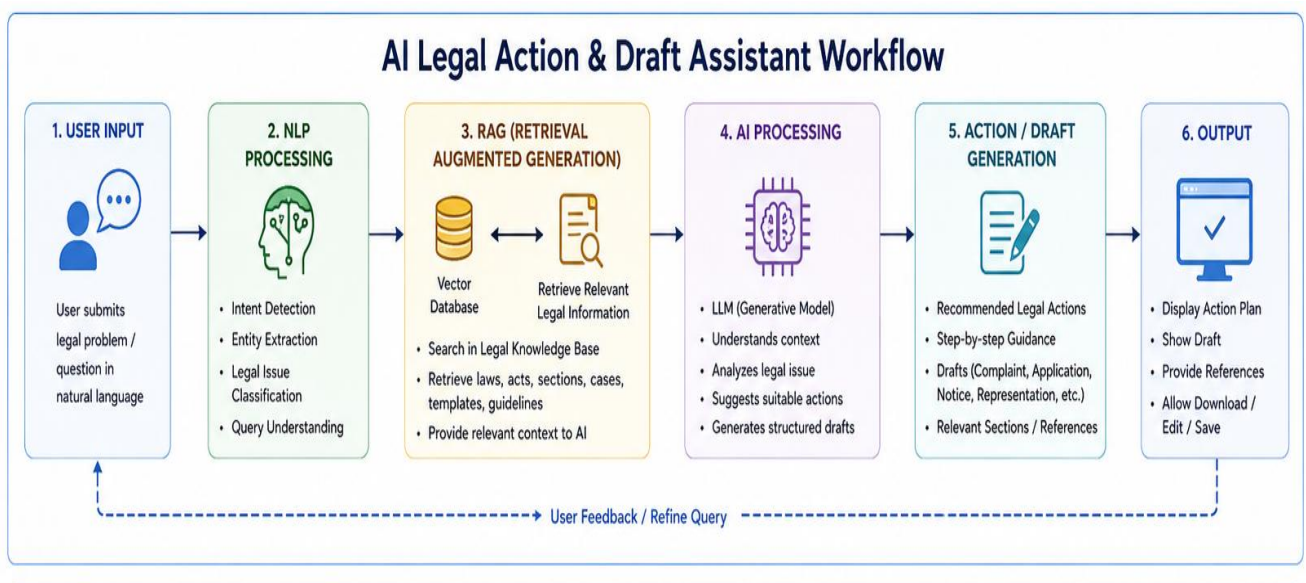
- **Average Response Time:** 4 seconds
- **Issue Identification Accuracy:** 85%
- **Action Recommendation Relevance:** 84%
- **System Availability:** 96%

Appendix B – Engineering Drawings / Schematics

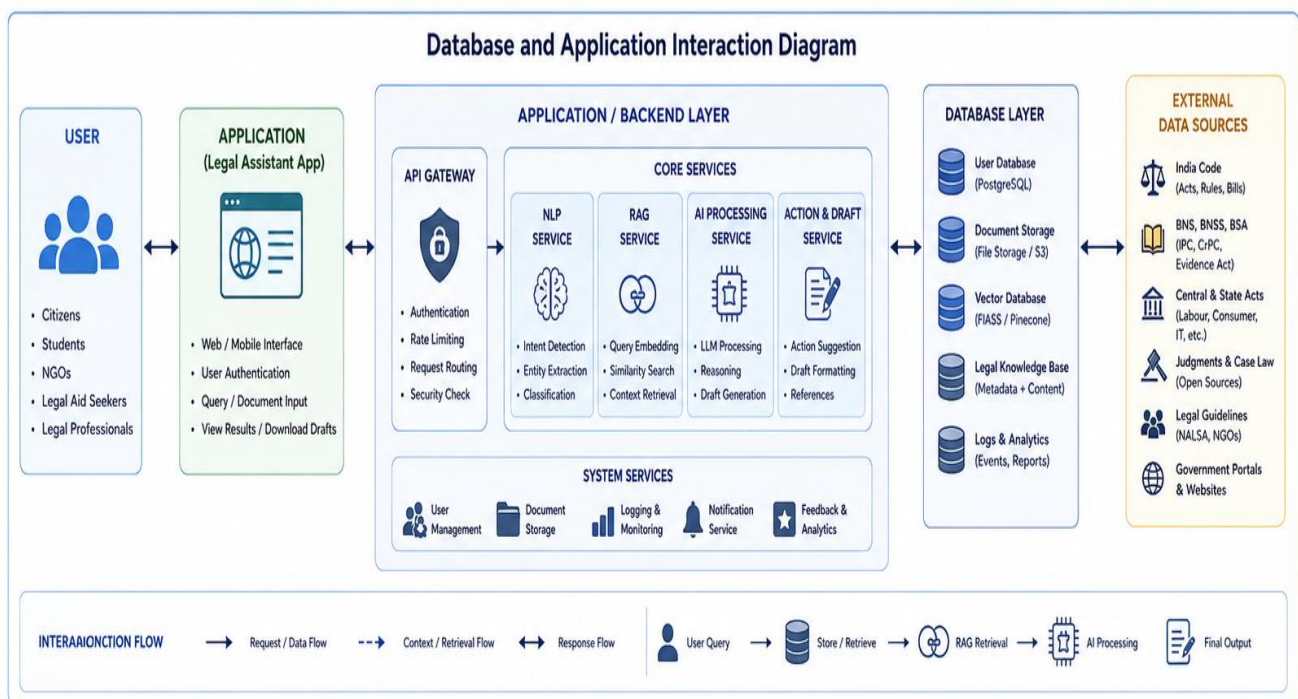
- Fig 2:NyayaMithra System Architecture



• Fig 3: AI Legal Action & Draft Assistant Workflow



• Fig 4: Database and Application Interaction Diagram



Appendix C – Source Code / Code Repository Information

Code:

```
import logging

from contextlib import asynccontextmanager
```

```

from fastapi import FastAPI, Request, status
from fastapi.middleware.cors import CORSMiddleware
from fastapi.responses import JSONResponse

from backend.config import ALLOWED_ORIGINS, BACKEND_HOST, BACKEND_PORT
from backend.api.routes import router as api_router
from backend.rag.retriever import get_retriever
from backend.services.ollama_service import get_ollama_service

# Configure clean logging
logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(levelname)s] %(name)s: %(message)s",
)
logger = logging.getLogger("nyayamithra.main")

@asynccontextmanager
async def lifespan(app: FastAPI):
    """
    Safe application lifecycle management.
    Performs non-blocking warmup checks for ChromaDB and Ollama without crashing if offline.
    """
    logger.info("Initializing NyayaMithra Legal AI System (Local-First)...")

    # Safe lazy probe of ChromaDB
    try:
        retriever = get_retriever()
        health = retriever.check_health()

```

```
    logger.info(f"Vector Store Status: {health.get('status')} ({health.get('total_chunks')} chunks indexed)")
```

```
except Exception as e:
```

```
    logger.warning(f"ChromaDB non-fatal warmup notice: {e}")
```

```
# Safe probe of Ollama
```

```
try:
```

```
    ollama = get_ollama_service()
```

```
    ollama_health = await ollama.check_health()
```

```
    if ollama_health.get("ollama_running"):
```

```
        logger.info(f'Ollama connected successfully. Active Model: {ollama_health.get('model')}')"
```

```
    else:
```

```
        logger.info(f'Ollama is offline or unreachable ({ollama_health.get('message')}). System is ready in fallback mode.")
```

```
except Exception as e:
```

```
    logger.warning(f'Ollama non-fatal probe notice: {e}')"
```

```
logger.info(f'NyayaMithra API ready at http://{BACKEND_HOST}:{BACKEND_PORT}')"
```

```
yield
```

```
logger.info("NyayaMithra shutting down.")
```

```
app = FastAPI(
```

```
    title="NyayaMithra API",
```

```
    description="Local-First Indian Legal Information & Legal-Document Drafting Assistant",
```

```
    version="1.0.0",
```

```
    lifespan=lifespan,
```

```
    docs_url="/docs",
```

```
    redoc_url="/redoc",
```

```
)
```

```

# CORS Configuration for local frontend environments
app.add_middleware(
    CORSMiddleware,
    allow_origins=ALLOWED_ORIGINS,
    allow_credentials=True,
    allow_methods=["GET", "POST", "PUT", "DELETE", "OPTIONS"],
    allow_headers=["*"],
)

@app.exception_handler(Exception)
async def global_exception_handler(request: Request, exc: Exception):
    """
    Global exception handler ensuring clean JSON errors without leaking internal stack traces.
    """
    logger.error(f"Unhandled error processing {request.method} {request.url.path}: {exc}",
exc_info=True)
    return JSONResponse(
        status_code=status.HTTP_500_INTERNAL_SERVER_ERROR,
        content={
            "error": True,
            "message": "An internal server error occurred while processing your legal request. Please
verify system status.",
        },
    )

# Mount API router
app.include_router(api_router)

@app.get("/")

```

```

async def root():
    return {
        "name": "NyayaMithra API",
        "description": "AI-powered Indian Legal Information & Legal-Document Drafting Assistant",
        "status": "online",
        "docs": "/docs",
        "health": "/api/health",
    }

if __name__ == "__main__":
    import uvicorn
    uvicorn.run(
        "backend.main:app",
        host=BACKEND_HOST,
        port=BACKEND_PORT,
        reload=True,
    )

```

Main technologies used:

- Python
- NLP
- Generative AI
- RAG

Appendix D – Datasheets

Not Applicable, as the project is primarily a software-based AI system and does not depend on specialized physical hardware components.

Appendix E – Experimental / Raw Data

The testing data consisted of sample legal documents used to evaluate the performance of the Smart Legal Document Intelligence module. The documents were processed to evaluate text extraction, document summarization, key information identification, and overall document processing performance.

Test Parameter	Result
Text Extraction Accuracy	94%
Summary Relevance	88%
Key Information Accuracy	86%
Response Time	6.8 sec
Document Processing	Successful
Legal Document Summarization	Successful

The experimental evaluation was performed using different sample legal documents containing clauses, legal terminology, dates, obligations, and other relevant information. The generated summaries and extracted information were compared with the original document content to assess their relevance and accuracy.

Detailed test cases, sample input documents, extracted text, generated summaries, and output screenshots may be included in this appendix as supporting evidence.

Appendix F – Ethics / Institutional Approval

The Smart Legal Document Intelligence module was designed with consideration for responsible use of Artificial Intelligence in the legal domain. Since legal documents may contain personal and confidential information, the system considers privacy and secure handling of uploaded documents during processing.

The generated summaries and extracted information are intended to support preliminary understanding of legal documents and are not intended to replace professional legal interpretation. Users should verify important information with the original document and seek assistance from a qualified legal professional whenever necessary.

Approval Reference: Not Applicable

Appendix G – Project Meeting / Progress Records

The development progress of the Smart Legal Document Intelligence module was monitored through regular discussions, implementation activities, testing, and refinement.

Stage	Activity
1	Requirement Analysis and identification of legal document-processing requirements
2	Literature Review and analysis of AI-based legal document systems
3	System Architecture and Document Processing Design
4	Text Extraction, Preprocessing and Document Chunking
5	Embedding, RAG and Relevant Information Retrieval
6	Generative AI-based Summarization and Key Information Extraction
7	Interface Integration and Output Display
8	Testing, Validation and Performance Evaluation
9	Final Implementation and Documentation

Appendix H – User/Industry Feedback

Where applicable, feedback from users, faculty, or project reviewers may be included.

Feedback Summary:

The Smart Legal Document Intelligence module was found useful for reducing the effort required to understand lengthy and complex legal documents. The ability to upload a document and obtain a simplified summary and important information was considered useful for preliminary document understanding.

Suggestions for further improvement include support for scanned documents through OCR, multilingual legal-document processing, improved extraction of important clauses, better document-level citations, and stronger validation of AI-generated information.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Publications: Not Applicable

Patents: Not Applicable

Competitions: Not Applicable

Product Outcome:

A working prototype of NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens was developed. The Smart Legal Document Intelligence module enables users to upload legal documents, extract textual content, retrieve relevant information, generate simplified summaries, and identify important legal information using AI-based document processing techniques.

Appendix J – Individual Contribution Evidence

Student	Contribution	Evidence
Team Member 1	AI Legal Advisor & Rights Navigator	Module 1 code, testing records and implementation screenshots
Team Member 2	Smart Legal Document Intelligence – document processing, text extraction, RAG/retrieval, summarization and key information extraction	Module 2 code, document-processing records, testing results and implementation screenshots
Team Member 3	AI Legal Action & Draft Assistant	Module 3 code, draft-generation records and testing screenshots

Evidence for My Module

The individual contribution for Module 2 – Smart Legal Document Intelligence can be demonstrated through the implemented document-processing code, sample legal documents used for testing, text extraction outputs, document chunking and retrieval results, generated summaries, key-information extraction results, test cases, and screenshots of the module interface.

This evidence demonstrates the development and testing work performed for the document intelligence component of the NyayaMithra system.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence / SO / Area	Location in This Report
Complex engineering problem formulation — Identification of difficulties in accessing legal information, understanding legal procedures, and preparing legal documents. SO1	Ch. 1, Ch. 2
Engineering design under constraints — Design of the GenAI, NLP, and RAG-based legal assistance system considering accuracy, response time, security, and usability. SO2	Ch. 3
Written/oral technical communication — Technical documentation, system explanation, results, and project presentation. SO3	Whole Report + Viva
Ethics and professional responsibility — Privacy, responsible AI usage, human review, and avoidance of misleading legal guidance. SO4	Ch. 3 (3.7)
Teamwork and leadership — Distribution of development, testing, documentation, and implementation responsibilities. SO5	Ch. 4 (4.8)
Experimentation, analysis & engineering judgment — Testing of response time, issue identification accuracy, action relevance, draft generation, and system availability. SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge — Learning Generative AI, NLP, RAG, prompt engineering, and AI-based document generation. SO7	Ch. 5 (5.3)
Standards — Application of information-security, data-protection, and web-application security considerations. SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact — Improving accessibility to preliminary legal assistance through a digital system and reducing dependence on manual drafting. SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation — Identification and mitigation of AI errors, data leakage, outdated legal information, and system failure. SO2/SO4	Ch. 3 (3.7)

Security considerations — Secure handling of user information, access control, and data protection. Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering — Integration of user interface, NLP, Generative AI, RAG, database, validation, and document generation. SO1/SO2	Ch. 3 (3.5)
Project planning and management — Requirement analysis, design, development, testing, and final implementation planning. SO5	Ch. 4 (4.8)
Individual contribution — Evidence of individual responsibilities and contributions to AI development, RAG, interface, testing, and documentation. SO1–SO7	Contribution Statement + Ch. 4 (4.8)